

SATVIKA EDA

Machine Learning Engineer

857-654-7302 | satvika2811@gmail.com | [linkedin.com/in/satvika-eda](https://www.linkedin.com/in/satvika-eda) | satvika-eda.github.io | Boston, MA

SUMMARY

Machine Learning Engineer with 3+ years of software development experience building and shipping high-performance production LLM systems, including fine-tuning, distillation, LLM inference optimization, analytics, and conversational AI evaluation. Experienced applying design patterns and scalable system architecture, collaborating closely with applied scientists on research to process data and scale ML models, and working in Agile environments to deliver reliable production deployments.

EXPERIENCE

CBRE

AI Engineer

Remote, USA

03/2026 – Present

- Developed a high-performance production LLM platform on Palantir AIP, scaling LLM inference across 30 parallel model calls with race condition handling on shared state, reducing manual report generation by 70% for 600 appraisers.
- Architected a conversational AI evaluation system validating 30 regulatory line items, implementing automated metrics and fallback logic to ensure reliable, high-performance LLM outputs with human-in-the-loop review, collapsing review time from hours to 30 seconds.
- Engineered React and TypeScript interfaces adopted by 600 appraisers across Wells Fargo, Bank of America, and Bank OZK, working in an Agile/Scrum environment to deliver production-grade features through iterative working sessions with business partners.

Dassault Systèmes

Machine Learning Engineer (Internship)

Boston, MA

08/2024 – 12/2024

- Led a 3-engineer team to build a production LLM platform on LangGraph, collaborating closely with applied scientists to prototype and scale a Llama-powered conversational AI assistant that cut engineering workflow time by 80%.
- Designed and evaluated an LLM inference pipeline with intent-classification routing across 6 specialized agents, investigating design approaches and optimizing inference to achieve 95% routing accuracy at scale.
- Built a scalable LLM data processing pipeline with FastAPI-served ETL and a RAG pipeline grounding conversational AI responses in domain-specific documentation, reducing generation time by 80% across multi-tenant environments.

JPMorgan Chase

Software Engineer

Hyderabad, India

03/2021 – 08/2023

- Co-invented U.S. Patent 12,223,309, a production ML platform in Java applying object-oriented design patterns and scalable architecture to predict deprecated SDK references across 100+ enterprise services, with CloudWatch, Grafana, and Splunk observability.
- Built OpenRewrite automation on Jenkins CI/CD enabling 90% automatic JDK upgrades across 100+ products, driving a 60% reduction in upgrade failure rate through production-grade reliability engineering.
- Designed and deployed 10+ microservices on Gaia cloud with a Light Switch utility automating service failover and restart workflows, reducing production downtime by 70% across distributed infrastructure.

PROJECTS

Multi-Agent Code Development System

- Built a production LLM fine-tuning and distillation pipeline, executing teacher-student distillation from Qwen2.5-7B to Qwen2.5-0.5B achieving 93% parameter reduction and 10x LLM inference speedup while preserving task performance through PPO-based RLHF fine-tuning.
- Designed an LLM training platform with 5 specialized agents for planning, implementation, debugging, and documentation, investigating design approaches and evaluating technical feasibility across 10+ fine-tuning iterations to continuously improve model performance.
- Automated A/B evaluation pipelines on Streamlit to assess LLM output quality, implementing data-backed feedback loops that reduced debugging time by 40% and enabled reliable conversational AI performance at scale.

LLM Evaluation Harness – MLOps Platform

- Designed an end-to-end MLOps platform with MLflow experiment tracking, a PostgreSQL model registry with candidate/staging/production promotion pipeline, and automated threshold-based alerting, benchmarking GPT-4o, Claude, and Mistral across RAGAS, DeepEval, and BERTScore with a live leaderboard dashboard.
- Built a scalable FastAPI backend with Redis/RQ for concurrent job orchestration, running evaluations across multiple LLM providers in parallel to reduce evaluation time by 70%, with real-time monitoring and automated metrics surfacing inference bottlenecks across workers.

Wealth Advisor Copilot – Enterprise Financial RAG System

- Designed a production LLM platform on LangGraph with a 6-stage RAG pipeline, pgvector-backed hybrid search, and Cohere Rerank over 50 SEC filings, achieving sub-3 second LLM inference latency with tenant-level data isolation.
- Integrated a 4-tier LLM guardrail system covering hallucination detection, PII redaction, and 3-level confidence scoring across 10 JWT-secured REST API endpoints with full audit logging.

EDUCATION

Northeastern University

Master of Science – Artificial Intelligence

Boston, MA

09/2023 – 12/2025

G Narayanamma Institute of Technology and Sciences

Bachelor of Technology – Computer Science

Hyderabad, India

08/2017 – 06/2021

SKILLS

Languages: Python, Java, C++, TypeScript, SQL, JavaScript

LLM Platform: Fine-Tuning, Distillation, LLM Inference, LLM Training, RLHF, Model Hosting, Inference Optimization, vLLM, SGLang, TensorRT, RLAI, PyTorch, TensorFlow, JAX, LLM Evaluation, Conversational AI, LangChain, LangGraph, LangSmith, RAG, Embeddings, Vector Databases (pgvector, ChromaDB), Prompt Engineering, LLM Guardrails, MLflow, LLMops

ML & Infrastructure: Agentic Systems, NLP, Deep Learning, Scikit-learn, XGBoost, LightGBM, SHAP, Statistical Modeling, Feature Engineering, MLOps, Docker, Kubernetes (EKS, GKE), Amazon Web Services (EC2, S3, SageMaker, Bedrock, CloudWatch), GCP, Vertex AI, FastAPI, REST APIs, PostgreSQL, Redis, CI/CD

Tools & Practices: Design Patterns, System Architecture, Reliability Engineering, Agile/Scrum, Neo4j, Git, Linux, Jenkins, Claude Code, Grafana, Splunk, Databricks, Palantir (AIP, Workshop, Ontology), React, PySpark, Snowflake, Terraform, GitHub Copilot, Jira